



AD23431 - STATISTICAL ANALYSIS AND COMPUTING**EXP 5: Switch, If then else, Looping statements**

1. A data analyst wants to apply different normalization techniques to a numeric vector based on user choice. Write an R program using switch() to perform:

- Min-Max Scaling
- Z-Score Normalization
- Decimal Scaling

The program should take a numeric vector as input and display the transformed output.

```
x <- c(10,20,30,40,50)
choice <- "minmax"
result <- switch(choice,
  minmax = (x - min(x))/(max(x) - min(x)),
  zscore = (x - mean(x))/sd(x),
  decimal = x/(10^nchar(max(x)))
)
result
```

Output:

```
[1] 0.00 0.25 0.50 0.75 1.00
```

2. In a classification task, a data scientist wants to calculate different evaluation metrics based on user selection. Write an R program using switch() to compute:

- accuracy
- precision
- recall

Use predefined values of TP, TN, FP, and FN and display the selected metric.

```
TP <- 50
TN <- 40
FP <- 5
```



```
FN <- 5
metric <- "accuracy"
result <- switch(metric,
  accuracy = (TP+TN)/(TP+TN+FP+FN),
  precision = TP/(TP+FP),
  recall = TP/(TP+FN)
)
result
```

Output:

[1] 0.9

3. Write an R program using if-else to check whether a given numeric value is an outlier based on the condition: A value is considered an outlier if it lies outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$. Assume $Q1$, $Q3$, and IQR values are given.

```
value <- 120
Q1 <- 50
Q3 <- 100
IQR <- Q3 - Q1
if(value < (Q1 - 1.5*IQR) | value > (Q3 + 1.5*IQR))
{
  print("Outlier")
}
else
{
  print("Not Outlier")
}
```

Output:

[1] "Not Outlier"



4. A retail company segments customers based on their annual spending:
Write an R program using if-else to categorize a customer.

- ≥ 100000 – Premium Customer
- 50000 to 99999 – Gold Customer
- 20000 to 49999 – Silver Customer
- < 20000 – Regular Customer

```
spending <- 65000
```

```
if(spending >= 100000)
{
  print("Premium Customer")
}
else if(spending >= 50000)
{
  print("Gold Customer")
}
else if(spending >= 20000)
{
  print("Silver Customer")
}
else
{
  print("Regular Customer")
}
```

Output:

```
[1] "Gold Customer"
```

5. In a binary classification model:

- If accuracy $\geq 90\%$ $\rightarrow$ "Excellent Model"
- 75%–89% $\rightarrow$ "Good Model"
- 60%–74% $\rightarrow$ "Average Model"
- Below 60% $\rightarrow$ "Poor Model"

Write an R program using if-else statements to interpret model performance.



```
accuracy <- 82
```

```
if(accuracy >= 90)
{
  print("Excellent Model")
}
else if(accuracy >= 75)
{
  print("Good Model")
}
else if(accuracy >= 60)
{
  print("Average Model")
}
else
{
  print("Poor Model")
}
```

Output:

```
[1] "Good Model"
```

6. Given a numeric vector representing daily sales for 10 days, write an R program using a for loop to calculate the mean without using the built-in mean() function.

```
sales <- c(100,120,130,150,170,200,220,210,190,180)
sum <- 0
for(i in sales)
{
  sum <- sum + i
}
mean_value <- sum/length(sales)
mean_value
```

Output:

```
[1] 167
```



7. Given a dataset column containing NA values, write an R program using a for loop to count the total number of missing values.

```
data <- c(10,20,NA,30,NA,40,50)
count <- 0
for(i in data)
{
  if(is.na(i))
  {
    count <- count + 1
  }
}
Count
```

Output:

```
[1] 2
```

8. Suppose a model improves its accuracy by 2% after each training iteration. Initial accuracy is 60%. Write an R program using a while loop that keeps increasing accuracy until it reaches or exceeds 85%. Display the number of iterations required.

```
accuracy <- 60
iteration <- 0

while(accuracy < 85)
{
  accuracy <- accuracy + 2
  iteration <- iteration + 1
}
iteration
accuracy
```

Output:

```
[1] 13
```

```
[1] 86
```